Exact nonconstant lower-chart control from two lines

November 13, 2026. Unsigned producer evidence [pcc:unverified].

Inputs and construction. Over $\mathbb{F}_{17}$ reuse $f = X^3 + 2X + 2$. Choose $L = 2X + 8$, $M = 4X + 6$. Lean checks

$$A = f + LM = X^3 + 8X^2 + 12X + 16, \quad B = L + M = 6X + 14 = 6H,\ H = X + 8.$$

The universal commutative-ring two-line identity and its exact specialization give

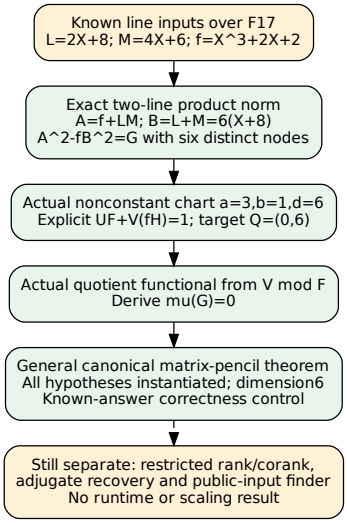
$$(f + LM)^2 - f(L + M)^2 = (f - L^2)(f - M^2),$$
$$A^2 - fB^2 = (X - 5)(X - 6)(X - 10)X(X - 3)(X - 13) = G.$$

All six nodes are distinct, and each supplied point $(5, 1), (6, 3), (10, 11), (0, 6), (3, 1), (13, 7)$ satisfies the cubic equation. At target $Q = (0, 6)$, $A(0) - 6B(0) = 0$ and $A(0) \neq 0$.

Explicit quotient data. Let $F = A$ and

$$U = 16X^3 + 11X^2 + 6X + 12, \qquad V = X^2 + 6X + 4.$$

The actual identity $UF + V(fH) = 1$ is checked. Degrees and monicity give the full chart $a = 3, b = 1, d = 6$, with B genuinely nonconstant. For $\mu(P) = [X^2]((PV) \bmod F)$, the supplied norm equation derives $\mu(G) = 0$. All hypotheses of the actual canonical matrix-pencil theorem are instantiated; the coordinate dimension is6.



Control scope, exact checks and next validation

Why this control. The earlier constant-B fixture did not exercise a positive-degree quotient modulo H. This algebraically constructed nonconstant chart covers that mechanism with supplied norm, inverse and branch data. It is a known-answer correctness control, with no random search or host-dependent runtime measurement. Exact equalities are independent of CPU contention. Restricted rank/corank, adjugate recovery and search complexity are not established by the current control. The supports are constructed with the witness; relation finding over an independently generated known-label online factor base is not tested.

Source and map. Module: `NormNonconstantControl.lean`.

Frozen source `cbfdaad42`. The map covers universal two-line norm; exact construction, cubic factors and prescribed norm; lower scaling and Bézout; branch/nonzero conditions; monicity/degrees; node distinctness and curve points; quotient target annihilation; coordinate dimension; and canonical matrix-pencil instantiation. Full declaration names are in the Markdown map. Twenty-two scoped interpretive records bind the reported declarations; all standing remains unsigned. The older NormRank source supplies the previous curve's nonsingularity and point-count checks.

Checks and failures. Lean4.35.0, unchanged pinned dependencies. Build004 passes3150 jobs without new-source warnings. Full tracked-source audit covers143 declarations, zero failing and zero additional trusted axioms. Separate fresh import-inclusive replay exits0; its completion receipt is retained. Only `propext`, `Classical.choice`, `Quot.sound` are admitted. Build001 left arithmetic side goals and dependent-index rewrite issues; actual degree-indexed families with a separate dimension proof resolve them. Build002 left nonzero6 over F17, discharged by kernel decide. Build003 passed; build004 removes unused new tactics. All logs are retained. No signed PCC verdict or kernel bypass is claimed.

Native identity. `IDC1h36c0f0c618539c35` checks universal `twoLineNorm` at32 statement-derived and32 extra-stream points through the unchanged maintained checker. The exact source pointer is to a universal CommRing theorem, not a finite-field-only equality. It does not execute Lean or establish finite-field/matrix/search claims. Its adapter's22 inherited tests pass, including tamper/statement-swapping controls. No timing claim.

Next. Actual arbitrary-pair bilinear/branch transfer, coordinate injectivity and corank; then guarded adjugate recovery and an explicit restricted control. Exceptional charts and a public-input moment finder with accounted $2^{o(n)}$ bit cost remain separate. Current data uses a supplied witness. Reachability through $W_{\mathrm{base}} + C(T)H(T)$ also remains open.

Sources and resources. Proposed two-line control, existing exact F17 fixture, actual matrix/functional source, pinned degree/monicity APIs. KB/conductor unavailable; bounded fallback, no priority claim for elementary line norms. No paid job, runtime/scaling campaign or budget override; monthly isolab budget exhausted. No benchmark/leaderboard/ecdb changes. Source-linked Markdown, editable DOT/SVG, diagram/report PDFs and raw evidence accompany this known-answer result.